

Second-Order Polarization Mode Dispersion: Impact on Analog and Digital Transmissions

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Abstract—Second-order polarization mode dispersion (PMD) is a major limitation to the transmission capabilities of analog systems and of high bit rate digital systems. Basically, the effect of second-order PMD is the same as that of chromatic dispersion. However, like all polarization effects in standard single-mode fibers, the effects of second-order PMD are stochastic, due to the random polarization mode coupling that occurs in such fibers. Hence, the effects of second-order PMD fluctuate with time. The purpose of this article is to present a description of the phenomenon and of its effects on analog and digital signals, to propose definitions, to discuss some orders of magnitudes and to present some numerical simulations and experimental results. Some general understanding of the phenomena of first-order PMD are assumed. The more mathematical developments are summarized in the Appendixes.

Index Terms—Optical fiber dispersion, optical fiber measurements, optical fiber polarization.

I. INTRODUCTION

DIGITAL communications systems are moving fast toward long distance, dense WDM, high bit rate optical links. Analog optical systems are also coming up into the backbone of cable TV distribution networks. This trend imposes strict requirements on the quality of optical cables, transmitters and network devices. Major limitations on the linearity and bandwidth of optical links are related to the spectral width of the optical source. Much effort has been made in chirp-free modulation techniques and equalization schemes, but nowadays bit rates are so high that the spectral content of the information stream itself is wide enough to be affected by residual dispersion in optical systems.

Polarization mode dispersion (PMD), which is due to the residual birefringence of a single mode optical fiber, breaks the degeneracy of the fundamental spatial mode into two orthogonal polarization modes, introducing a time delay between them that gives rise to signal distortion [1]–[3]. Because of the constantly varying distributed environmental conditions of optical cables, polarization effects are stochastic due to random mode

coupling which occurs in standard fibers [4]. Hence, unlike chromatic dispersion, PMD effects are constantly varying and cannot be compensated by usual equalization techniques such as chirped gratings and dispersion shifted fibers. Therefore, although small, PMD cumulate in long distance links and becomes relevant in long distance amplified digital optical links [2]. Moreover, because birefringence effects are wavelength dependent, PMD is also wavelength dependent, and this feature, which is referred to as second-order PMD, is similar to the chromatic dispersion, and due to its stochastic character, becomes the major limiting factor to the bandwidth and linearity of optical systems [1].

The purpose of this paper is to present a description of the phenomenon and of its effects on analog and digital optical systems, to propose definitions, and to discuss some order of magnitude effects. Definition of second-order PMD is made in Section II, while its effects and statistical properties are presented in Sections II and III, together with some numerical simulations. Experimental results are presented in Section IV, measurement implications and standardization problems are discussed in Sections V and VI. Finally, polarization dependent loss effects are briefly discussed in Section VII. The more mathematical developments are summarized in the Appendixes.

II. DISTORTION IN ANALOG SYSTEMS AND DEFINITION OF SECOND-ORDER PMD

Let us consider the simplest analog signal, a pure sinusoidal intensity modulated light signal produced by a laser

$$I_{in}(t) = I_0[1 + \alpha \sin(\omega_m t)] \quad (1)$$

where α is the modulation depth and ω_m the modulation frequency. If such a signal propagates through a fiber several effects can take place. First, there is a global attenuation. Next, there is a global time delay (i.e., a phase shift in the sinus). These two phenomena are irrelevant for our present discussion, since they do not introduce any distortion. However, if either the attenuation or the time delay varies in time (for any reason), this would produce harmonics: the output signal would consist of the input sinus plus a signal at twice the frequency (and possibly higher order harmonics). An example of variable attenuation is polarization dependent losses (PDL). Examples of the second cause of distortion are the combination of laser chirp with chromatic dispersion and/or second-order PMD. In the following we concentrate on the latter.

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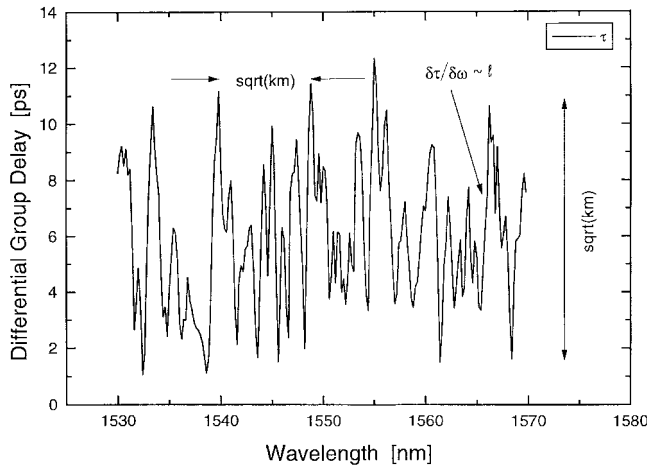


Fig. 1. Typical differential group delay (DGD) curve as a function of wavelength. Second-order PMD is related to the dependence of the DGD on the optical frequency (however, a complete description has to take into account the complete three-dimensional (3-D) principal polarization state vector). When the fiber length ℓ increases, the DGD's increase as $\sqrt{\ell}$, as indicate on the right-hand side (RHS) of the figure. Simultaneously, the oscillations' period decrease with the inverse of $\sqrt{\ell}$. This explains that the second-order PMD increases linearly with the fiber length.

Intuitively, the phenomena of second-order PMD happens as follows. The intensity modulation of the laser produces an optical frequency modulation (this is called the laser chirp). Now, since the time delay due to first-order PMD is very wavelength sensitive, this frequency modulation implies in turn a modulation of the propagation time. Hence a phase modulation of the output signal. This phase modulation produces harmonics. If a combination of harmonics (additions and subtractions of modulation frequencies) matches another modulation frequency (i.e., another transmission channel), then this channel is perturbed. More precisely, because of the laser chirp, the instantaneous laser optical frequency $\omega(t)$ is also sinusoidally modulated

$$\omega(t) = \omega_0 + \gamma \sin(\omega_m t) \quad (2)$$

where ω_0 is the central optical frequency and γ the effective laser chirp ($\gamma = \eta \delta I / 2$ with η the chirp coefficient in [GHz/mA] and δI the peak to peak current modulation in [mA]). Now, assuming that the fiber has some nonvanishing random birefringence, it has to be treated as a two-polarization-mode fiber.¹ These two polarization modes are called the principal polarization states and are known to be very wavelength sensitive [3], [5], as illustrated on Fig. 1. For each optical frequency, the two principal states are orthogonal and travel at different group velocities, provided PDL is negligible. Hence, each optical frequency sees a different pair of polarization modes and undergoes different time delays. Since, because of the laser chirp, the optical frequency varies in time, the time delays do also vary in time. Moreover, the relative power launched into each of the two principal polarization modes does also vary in time. Hence, the output signal is distorted. This phenomenon is called second-order PMD.

¹ We assume that the coherence time of the laser is much larger than the differential group delay (DGD), a very good approximation in most practical cases.

In CATV systems, the signal is not merely a sinus, but a combination of tens of signals at different frequencies, each with its own modulation. However, the basic physical phenomena is the same. For instance, a combination of three frequencies may interfere with a fourth one, producing unwanted diaphonies.

In order to be more precise and specific we need to introduce some notations. Recall that the principal polarization modes can be defined as the fastest and slowest polarization modes for light pulses with a coherence time much larger than the PMD delay $\Delta\tau$ of the fiber [3], [5]. In the absence of polarization dependent losses, these two modes are mutually orthogonal and can be represented on the Poincaré sphere by two opposite points in directions $\pm\vec{\Omega}$, respectively. By convention the length of the principal polarization vectors equals the DGD $|\vec{\Omega}| = \delta\tau$. Hence, $\delta\tau(\omega)$ denotes the DGD due to PMD. Furthermore, let $\tau(\omega)$ denote the mean time delay (mean over the two polarization modes). Both depends on the optical frequency ω . Accordingly, the time delay of the fast and slow principal polarization modes are $\tau(\omega) \mp \frac{1}{2}\delta\tau(\omega)$, respectively. Each of these two modes are subject to the effect of chromatic dispersion, corresponding to the above time delays (see Appendix A). In addition to this, the ratio intensities of the fast and slow modes is modulated because the principal polarization state depends also on the optical frequency

$$I_{\text{out}}(t) = I_{\text{out,fast}}(t) \frac{1 + \hat{\Omega}[\omega(t)]\vec{m}_{\text{laser}}}{2} + I_{\text{out,slow}}(t) \frac{1 - \hat{\Omega}[\omega(t)]\vec{m}_{\text{laser}}}{2} \quad (3)$$

where $\hat{\Omega} \equiv \vec{\Omega}/|\vec{\Omega}|$ is the unit vector in the direction $\vec{\Omega}[\omega(t)]$. Note that all polarization states are represented using the Poincaré sphere. Hence, the intensity I of a polarized signal $\vec{m}_{\text{laser}}$ passing through a polarizer of direction $\vec{\Omega}$ values $I \cos^2(\theta)$ where θ is twice the angle between the Poincaré vectors $\vec{m}_{\text{laser}}$ and $\vec{\Omega}$, or equivalently in case of linear polarization the angle between the two polarization directions. This law is named Malus law and is represented on the Poincaré sphere by $(1 + \hat{\Omega}\vec{m}_{\text{laser}})/2 = \cos^2(\theta)$.

Appendix A develops this expression of $I_{\text{out}}(t)$ and finds the RF power at twice the modulation frequency ω_m relative to the RF power at the fundamental modulation frequency

$$\eta^{(2)} \equiv \frac{(\text{RFat } 2\omega_m)_{\text{out}}}{(\text{RFat } \omega_m)_{\text{in}}} = (\gamma\omega_m)^2 \left[\tau' - \frac{\vec{\Omega}_\omega(\omega_0)\vec{m}_{\text{laser}}}{2} \right]^2 \quad (4)$$

where $\tau' \equiv [d\tau/d\omega] = [\lambda^2\ell/2\pi c]D$ represents the chromatic dispersion contribution, with D the chromatic dispersion coefficient in ps/km.nm , c is the speed of light, λ the wavelength and ℓ the fiber length. In (4), we introduced the derivative of the principal polarization state vector with respect to the optical frequency

$$\vec{\Omega}_\omega \equiv \frac{d\vec{\Omega}}{d\omega}. \quad (5)$$

We like to emphasize that second-order PMD is represented by a three-component vector $\vec{\Omega}_\omega$, as this point is often mistaken in the literature.

Accordingly, we see that the effect of second-order PMD can either add or subtract to chromatic dispersion, depending on the sign of $\vec{\Omega}_\omega \vec{m}_{\text{laser}}$. This result was first found by Poole and Darcie in [1], though using a more complicated derivation.

In practice the laser polarization state is unknown. Moreover, the orientation of the second-order PMD vector $\vec{\Omega}_\omega$ is fluctuating. Hence it is useful to consider $\bar{\eta}^{(2)}$, the average of $\eta^{(2)}$ over the relative orientation of the two vectors that appear in (4)

$$\bar{\eta}^{(2)} = (\gamma\omega_m)^2 \left[(\tau')^2 + \frac{\vec{\Omega}_\omega^2(\omega_0)}{12} \right]. \quad (6)$$

Section IV presents results about the fiber length dependence and the statistics of $\bar{\eta}^{(2)}$. But before, we like to propose a definition for a second-order PMD coefficient.

Second-order PMD is described by a three-component vector $\vec{\Omega}_\omega$. The mean of this vector is zero, hence we propose to characterize second-order PMD by its rms value. Moreover, since the effect of second-order PMD in (6) adds to chromatic dispersion, we propose to define the second-order PMD coefficient in such a way that it has the same units as chromatic dispersion

$$\boxed{\text{PMD}_{\text{coeff}}^{\text{second-order}} = \frac{2\pi c}{\lambda^2 \ell} \sqrt{\langle \vec{\Omega}_\omega^2(\omega) \rangle} \left[\frac{\text{ps}}{\text{nm km}} \right]} \quad (7)$$

where $\langle \dots \rangle$ represents the statistical mean. In practice one expects this mean value to be the same as a wavelength average, similarly to the case of first-order PMD. Note that *this definition is only valid when the fiber is long* compared to the mean polarization mode coupling length h [3].

It is convenient to define a factor u whose value almost equals one at telecom wavelengths when expressed in units of 1/(nm.ps)

$$u \equiv \frac{2\pi c}{\lambda^2} \left[\frac{1}{\text{nm ps}} \right] \approx 1 (\text{for practical purposes}). \quad (8)$$

So in a compact form the second-order PMD coefficient reads

$$\text{PMD}_{\text{coeff}}^{\text{second-order}} = \frac{u}{\ell} \sqrt{\langle \vec{\Omega}_\omega^2(\omega) \rangle} \left[\frac{\text{ps}}{\text{nm km}} \right]. \quad (9)$$

Fig. 1 presents the DGD (i.e., the time delay between the two principal polarization modes) in function of wavelength. Intuitively, second-order PMD is related to the slope of this curve. However, as illustrated by relations (3) and (4) above, a more complete description of the phenomenon involves also the full wavelength dependence of the principal states, not only the DGD's. Therefore, it is important to define second-order PMD as above and not as the derivative of the curve presented in Fig. 1.

The second-order PMD vector $\vec{\Omega}_\omega$ is related to the derivative of the DGD with respect to frequency $\delta\tau' \equiv (\partial\delta\tau)/(\partial\omega)$ as follows:

$$\vec{\Omega}_\omega^2 = [\partial_\omega(\hat{\Omega}\delta\tau)]^2 = (\delta\tau')^2 + \delta\tau^2 \left(\frac{\partial\hat{\Omega}}{\partial\omega} \right)^2. \quad (10)$$

Numerical evidence shows that the second term in this decomposition is about eight times larger than the first term (see

Section III). Accordingly, second-order PMD should not be confused with the wavelength dependence of the DGD, as this contributed only to about 1/9th of the total second-order PMD effect!

For completeness, let us recall the root-mean-square (rms) definition of the first-order PMD coefficient [3]

$$\text{PMD}_{\text{coeff}}^{\text{first-order}} = \frac{1}{\sqrt{\ell}} \sqrt{\langle \vec{\Omega}^2(\omega) \rangle} = \frac{\Delta\tau}{\sqrt{\ell}}. \quad (11)$$

The contribution of second-order PMD to composite second-order (CSO) can also be expressed as a function of the second-order PMD vector $\vec{\Omega}_\omega$

$$\text{CSO} = 10 \log [4N\eta^{(2)}] = 10 \log \left(N \frac{\gamma^2 \omega^2}{3} \vec{\Omega}_\omega^2 \right) \quad (12)$$

where N is the product count, i.e., the number of frequency products in the considered RF-channel, and where the factor four takes into account that this relation is valid for the non-degenerate case, when the various frequencies that combine to produce the undesired frequency are all different.

III. PULSE BROADENING DUE TO SECOND-ORDER PMD

In this section, we compute the pulse spreading due to first-order chromatic dispersion and to first- and second-order PMD for a Fourier transform limited Gaussian input pulse of spread $\sigma_{\text{in}} = 1/\Delta\omega$

$$E_{\text{in}}(t) = A_0 \cdot e^{-t^2 \Delta\omega^2/4} \cdot e^{-i\omega_0 t} \cdot \psi_{\text{in}} \quad (13)$$

where ψ_{in} represents the polarization state of the input pulse. As chromatic dispersion and second-order PMD assume frequency dependent delays, the computation has to be performed in the frequency domain

$$E_{\text{in}}(\omega) = N \cdot e^{-(\omega-\omega_0)^2/\Delta\omega^2} \cdot \psi_{\text{in}} \quad (14)$$

where N is a normalization constant. Taking into account chromatic dispersion and PMD, the pulse at the output of the fiber can be described by

$$E_{\text{out}}(\omega) = N \cdot e^{-(\omega-\omega_0)^2/\Delta\omega^2} \cdot \left\{ e^{i[\beta(\omega)\ell - \int_{\omega_0}^{\omega} \delta\tau(\tilde{\omega})d\tilde{\omega}/2]} < \psi_{\text{in}}^+(\omega) | \psi_{\text{in}} > \psi_{\text{out}}^+(\omega) + e^{i[\beta(\omega)\ell + \int_{\omega_0}^{\omega} \delta\tau(\tilde{\omega})d\tilde{\omega}/2]} < \psi_{\text{in}}^-(\omega) | \psi_{\text{in}} > \psi_{\text{out}}^-(\omega) \right\} \quad (15)$$

where $\psi_{\text{in}}^\pm(\omega)$ and $\psi_{\text{out}}^\pm(\omega)$ denote the fast and slow principal polarization state Jones vectors at frequency ω at the input and output of the fiber, $\beta(\omega)$ is the propagation constant, ℓ the fiber length and $\delta\tau(\omega)$ the DGD. Assuming $\Delta\omega$ small, we develop β and $\delta\tau$ in $\delta\omega \equiv (\omega - \omega_0)$

$$\begin{aligned} \beta(\omega)\ell &= \beta_0\ell + \delta\omega\beta'_0\ell + \frac{\delta\omega^2}{2}\beta''_0\ell + O(\delta\omega^3) \\ &= \phi_0 + \delta\omega \cdot t_0 - \frac{\delta\omega^2}{2}\tau' + O(\delta\omega^3) \end{aligned} \quad (16)$$

$$\int_{\omega_0}^{\omega} \delta\tau(\tilde{\omega})d\tilde{\omega} = \delta\omega \cdot \delta\tau_0 + \delta\omega^2 \cdot \frac{\delta\tau'_0}{2} + O(\delta\omega^3) \quad (17)$$

TABLE I

EVALUATION OF THE VARIOUS TERMS CONTRIBUTING TO THE PULSE SPREADING DUE TO (FIRST-ORDER) CHROMATIC DISPERSION AND TO FIRST AND SECOND-ORDER PMD. THE SECOND COLUMN PRESENTS THE MEAN VALUES OVER ALL POSSIBLE INPUT POLARIZATION STATES. IN CASE THIS MEAN VANISHES, THE RMS VALUE IS DISPLAYED WITH $\pm$ SIGN. THE THIRD COLUMN PRESENTS THE WAVELENGTH AVERAGE. THESE ARE PROPORTIONAL TO VARIOUS COMBINATIONS OF THE FIRST-ORDER PMD $\Delta\tau$, THE CHROMATIC DISPERSION τ' AND THE SOURCE BANDWIDTH $\Delta\omega$. THE PROPORTIONALITY COEFFICIENT IS EITHER THE THEORETICAL ONE, WHENEVER AVAILABLE, OR DEDUCED FROM NUMERICAL SIMULATIONS. THE LAST COLUMN PRESENTS ORDERS OF MAGNITUDES FOR TWO CASES: A 100-KM LONG FIBER WITH A PMD $\Delta\tau = 10$ ps AND A 50-KM LONG FIBER WITH $\Delta\tau = 30$ ps. IN BOTH CASES A STANDARD FIBER IN THE THIRD TELECOM WINDOW (i.e., 1550 nm) IS ASSUMED WITH $\Delta\omega = 0.1$ (1/ps) = $(100/2\pi)$ GHz ≈ 16 GHz AND A CHROMATIC DISPERSION $D = 17$ ps/km.nm)

$\sigma_{out}^2 =$	Average (or $\pm$ RMS) over the input polarization	Average over wavelength	numerical simulation $\Delta\tau = 10$ $\tau' = 1700$ [ps ²]	$\Delta\tau = 30$ $\tau' = 800$ [ps ²]
$+\sigma_{in}^2$	σ_{in}^2	σ_{in}^2	100	100
$+\frac{1}{4}\delta\tau^2(1-m_3^2)$	$\frac{1}{6}\delta\tau^2$	$\frac{1}{6}\Delta\tau^2$	16.6	150
$+\frac{1}{4}\tau'^2\Delta\omega^2$	$+\frac{1}{4}\tau'^2\Delta\omega^2$	$+\frac{1}{4}\tau'^2\Delta\omega^2$	7225	1600
$+(\alpha\delta\tau m_1 + \frac{1}{2}\delta\tau' m_3)\tau'\Delta\omega^2$	$\pm\frac{1}{\sqrt{12}}\Omega_\omega\tau'\Delta\omega^2$	$\pm\frac{1}{6}\Delta\tau^2\tau'\Delta\omega^2$	± 283	± 1200
$+\frac{1}{4}\Omega_\omega^2\Delta\omega^2$	$+\frac{1}{4}\Omega_\omega^2\Delta\omega^2$	$\frac{1}{12}\Delta\tau^4\Delta\omega^2$	8.3	675
$-\frac{3}{4}\alpha^2\delta\tau^2(1-m_3^2)\Delta\omega^2$	$-\frac{3}{2}\alpha^2\delta\tau^2\Delta\omega^2$	$-0.037\Delta\tau^4\Delta\omega^2$	-3.7	-300
$-\frac{3}{4}\alpha\delta\tau\delta\tau'm_3m_1\Delta\omega^2$	$\pm\frac{1}{4}\sqrt{\frac{3}{5}}\alpha\delta\tau\delta\tau'\Delta\omega^2$	$0 \pm 0.016\Delta\tau^4\Delta\omega^2$	± 1.6	± 130
$+\frac{1}{8}\alpha\delta\tau^3m_3m_2\Delta\omega^2$	$\pm\frac{1}{8\sqrt{15}}\alpha\delta\tau^3\Delta\omega^2$	$\pm 0.009\Delta\tau^4\Delta\omega^2$	± 0.9	± 73

where τ' is related to the chromatic dispersion coefficient D as in the previous section $D = (2\pi c)/(\lambda^2 \ell)\tau'$ [ps/nm.km]. The principal polarization state Jones vector should also be developed in $\delta\omega$. For compactness, we drop the indexes "in" and "out" as this has no influence on the pulse spread. Since both Jones vectors are normalized and mutually orthogonal for all ω , their first-order development reads, with proper choice of irrelevant phases

$$\psi^+(\omega) = \frac{\psi_0^+ + \delta\omega\alpha\psi_0^-}{\sqrt{1 + \delta\omega^2 \cdot \alpha^2}} + O(\delta\omega^2) \quad (18)$$

$$\psi^-(\omega) = \frac{\psi_0^- - \delta\omega\alpha\psi_0^+}{\sqrt{1 + \delta\omega^2 \cdot \alpha^2}} + O(\delta\omega^2) \quad (19)$$

where in this section, α describes the rate of change of direction of the principal polarization vector $4\alpha^2 = [\partial_\omega(\vec{\Omega}/\delta\tau)]^2$. In Appendix B, we show that the coefficient α is the same for both the input and the output principal polarization states $\alpha_{in} = \alpha_{out}$. Note that we normalize $\psi^\pm(\omega)$, keeping terms up to order $\delta\omega^2$ as in the developments (16) and (17). For mathematical completeness, one should include in (19) higher order derivatives, however we neglect them as they would physically correspond to third-order PMD.

In Appendix B the spread of the outcoming pulse

$$\sigma_{out}^2 = \frac{\int t^2 |E_{out}(t)|^2 dt}{\int |E_{out}(t)|^2 dt} - \left[\frac{\left(\int t |E_{out}(t)|^2 dt \right)^2}{\int |E_{out}(t)|^2 dt} \right] \quad (20)$$

is computed. Table I summarizes the results, together with some orders of magnitude. $\vec{m}$ denoting the input polarization state on the Poincaré sphere with $\vec{m} = (m_1\delta\omega, 0, m_3) + O(\delta\omega^2)$ the principal polarization state at ω_0 .

In presence of chromatic dispersion, the dominant new term for pulse spreading is the fourth one $\pm\frac{1}{6}\Delta\tau^2\tau'\Delta\omega^2$, which involves the product of the chromatic dispersion and second-order PMD [2]. Note that this contribution may be positive or negative. If chromatic dispersion is negligible, e.g., a standard

fiber in the second telecom window or in the third window with chromatic dispersion compensation it is the fifth term $\frac{1}{12}\Delta\tau^4\Delta\omega^2$ which dominates. Let us emphasize the second-order PMD effects can be as large as first-order effects, and even larger when combined with large chromatic dispersion [2]. Finally, we note that the dominant terms, fourth and fifth, both involve the second-order PMD vector $\vec{\Omega}_\omega$ which characterizes also the distortion of analog transmissions [see (6)]. This supports our definition (7) of the second PMD coefficient. The relation between $\vec{\Omega}_\omega$ and the derivative of the DGD $\delta\tau'$ and the rate or rotation of the principal polarization state α reads [see (10)] $\Omega_\omega^2 = 4\alpha^2 \cdot \delta\tau^2 + (\delta\tau')^2$. From the orders of magnitude given in Table I, one may see that the first term, involving the DGD and the rate of rotation of the principal polarization state, contributes (about eight times) more to second-order PMD than the derivation of the DGD. This is somewhat surprising and differs strongly from a common confusion in the literature where second-order PMD is sometimes identified with the wavelength dependence of the DGD.

Finally, let us briefly consider third-order PMD. Indeed, a mathematically proper development of the output field (15) should include all terms that contribute to the spread σ_{out}^2 up to order $\delta\omega^2$ and this involves higher order derivatives of the principal polarization vector $\vec{\Omega}$. For example, we have numerically investigated the magnitudes of the following terms: $\text{Mean}(\delta\tau'') \equiv \text{Mean}[\partial^2\delta\tau/\partial\omega^2] \approx 0 \pm 0.2\Delta\tau^3$, and $\text{Mean}[\partial^2(\vec{\Omega}/\delta\tau)/\partial\omega^2]^2 \approx 1.6\Delta\tau^4$. These mean values are not small, though their contribution to σ_{out} should be multiplied with relatively small coefficients coming from the Taylor expansions.

IV. STATISTICS OF SECOND-ORDER PMD AND FIBER LENGTH DEPENDENCE

In the two previous sections, we introduced the quantity $\vec{\Omega}_\omega$ that characterizes second-order PMD. In this section we summarize the main results about the statistics of this vector

and its fiber length dependence. The details of the derivations can be found in Appendixes C and D.

The basic equation we need is the dynamical equation for PMD [6], [7] which relates the principal polarization states to the fiber birefringence and length

$$\frac{d\vec{\Omega}(z, \omega)}{dz} = \vec{\beta}(z) + \omega \vec{\beta}(z) \wedge \vec{\Omega}(z, \omega) \quad (21)$$

where $\vec{\beta}(z)$ represents the local modal birefringence in [ps/km] at position z along the fiber.² It fluctuates along the fiber with zero mean and correlation length h . By taking the derivative of this equation, we obtain the dynamical equation for second-order PMD

$$\frac{d\vec{\Omega}_\omega}{dz} = \vec{\beta}(z) \wedge \vec{\Omega}(z, \omega) + \omega \vec{\beta}(z) \wedge \vec{\Omega}_\omega(z, \omega). \quad (22)$$

The general solution to this equation is not known. However, for fibers much longer than the mean polarization mode coupling length h , the random birefringence can be described as a Markovian stochastic process and the dynamical equation can be solved (see Appendix E). Recall that the mean polarization mode coupling length h can equivalently be defined as the length after which the polarization state is decorrelated from the initial polarization state, or as the correlation length of the fluctuating birefringence of the fiber [3]. It is thus a parameter characterizing a statistical property of the fiber.

The main consequences of the dynamical (22) are the following.

A. Fiber Length Dependence

We are interested in the magnitude of the vector $\vec{\Omega}_\omega$, in particular in its square $\vec{\Omega}_\omega^2$. In Appendix C, using the dynamical equation for second-order PMD (22), we derive its fiber length dependence

$$\langle \vec{\Omega}_\omega^2 \rangle = \frac{B^4 h^4}{3} \left[\frac{\ell^2}{h^2} - \frac{5\ell}{2h} + \frac{17}{8} - 2 \cdot \left(\frac{\ell}{h} + 1 \right) e^{-(2\ell/h)} - \frac{1}{8} e^{-(4\ell/h)} \right] \quad (23)$$

where $B^2 = \langle \vec{\beta}^2 \rangle$ is the rms modal birefringence, h the mean polarization mode coupling length and ℓ the fiber length. This fiber length dependence is illustrated in Fig. 2. Accordingly, second-order PMD increases only very slowly for short fibers, but increases linearly for fibers much longer than h (in practice longer than a few kilometers).

The two important cases where ℓ is large compared to h or where ℓ is negligible compared to h , follow from the limits:

$$\lim_{(\ell/h) \rightarrow \infty} \langle \vec{\Omega}_\omega^2 \rangle = \frac{B^4 h^4}{3} \left(\frac{\ell}{h} \right)^2 + 0 \left(\frac{\ell}{h} \right) \quad (24)$$

$$\lim_{(\ell/h) \rightarrow 0} \langle \vec{\Omega}_\omega^2 \rangle = \frac{4}{15} \frac{B^4 h^4}{3} \left(\frac{\ell}{h} \right)^5 + 0 \left(\frac{\ell}{h} \right)^6. \quad (25)$$

²Note that henceforth $\vec{\beta}$ denotes the modal birefringence, whereas in Section III, it denoted the propagation constant.

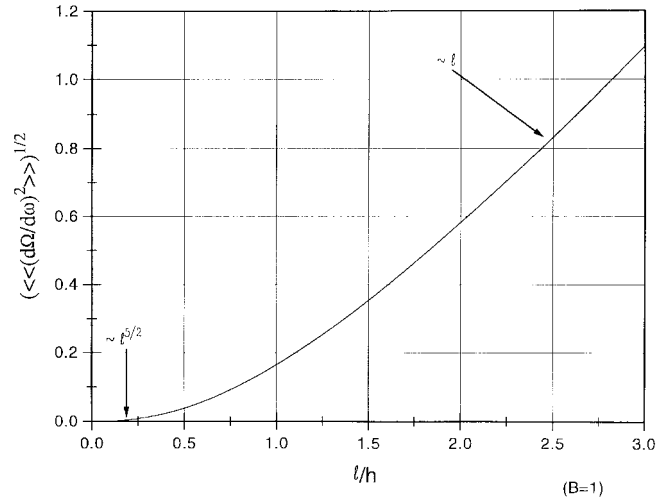


Fig. 2. Fiber length ℓ dependence of second-order PMD [see (23)].

Hence, for long fibers, the second-order PMD coefficient (7) relates to the fiber parameters as follows:

$$\lim_{(\ell/h) \rightarrow \infty} \text{PMD}_{\text{coeff}}^{\text{second-order}} = \frac{uB^2 h}{\sqrt{3}}. \quad (26)$$

Some intuitions about the linear dependence of second-order PMD on fiber length ℓ can be gained as follows. First, recall that first-order PMD increases with the square root of the fiber length. This is due to the diffusive nature of first-order PMD. Hence, the vertical scale in Fig. 1 expands with $\sqrt{\ell}$. Next, consider the horizontal scale of this figure. When the fiber gets longer, more and more frequencies appear on this curve $\text{DGD}(\lambda)$. Hence, more and more structures appear in Fig. 1. This corresponds to a contraction of the horizontal scale, somewhat like an accordion that closes. This contraction goes also with the square root of the fiber length $\sqrt{\ell}$. Hence, the slope of the $\text{DGD}(\lambda)$ curve increases linearly with ℓ . Likewise the derivation of each component of the PMD vector $\vec{\Omega}$ increase linearly with ℓ .

Note that the linear fiber length dependence of second-order PMD implies that the distortion parameter $\eta^{(2)}$ defined by (4) increases with the square of the fiber length.

B. Case of Long Fibers: Relation to First-Order PMD

As we have seen in the previous section, in the case of fibers long with respect to the mean coupling length h , second-order PMD increases linearly with the fiber length ℓ , contrary to first-order PMD which increases with the square root $\sqrt{\ell}$. This leads to the important result that for long fibers first and second-order PMD are connected by the following simple relation:

$$\langle \vec{\Omega}_\omega^2 \rangle = \frac{1}{3} \left(\langle \vec{\Omega}^2 \rangle \right)^2 \quad (27)$$

$$\langle \vec{\Omega}_\omega^4 \rangle = \left[\frac{5}{3} \left(\langle \vec{\Omega}^2 \rangle \right)^2 \right]^2. \quad (28)$$

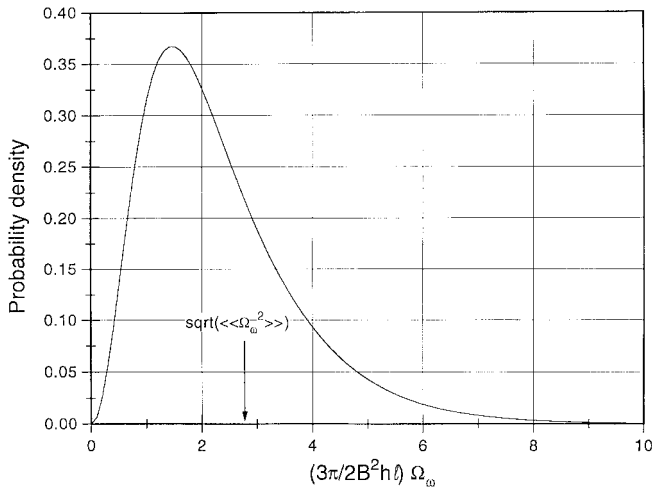


Fig. 3. Probability density distribution of second-order PMD [see (30)].

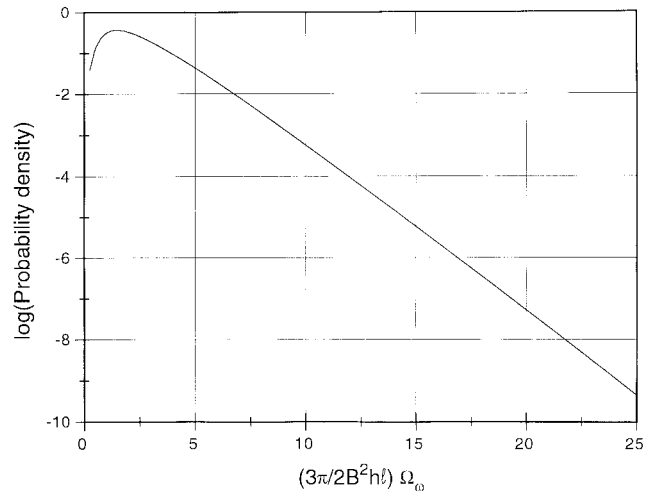


Fig. 4. Same as Fig. 3, but on a logarithmic scale.

Hence, adopting the rms definition of first-order PMD [3], we have [8]

$$\text{PMD}_{\text{coeff}}^{\text{second-order}} = \frac{u}{\sqrt{3}} (\text{PMD}_{\text{coeff}}^{\text{first-order}})^2 \approx \frac{\Delta\tau^2}{\sqrt{3}}. \quad (29)$$

For example, if $\text{PMD}_{\text{coeff}}^{\text{first-order}} = 0.5 \text{ ps}/\sqrt{\text{km}}$, then one has $\text{PMD}_{\text{coeff}}^{\text{second-order}} \approx 0.15 \text{ ps}/\text{nm.km}$.

C. Case of Long Fibers: Probability Distribution

In the case of long fibers, the probability distribution of the magnitude $\Omega_\omega = |\vec{\Omega}_\omega|$ can be derived from the dynamical equation for second-order PMD (22), see Appendix D

$$\rho_\ell(\Omega_\omega) = \frac{2\xi^2\Omega_\omega \sinh(\xi\Omega_\omega/\ell)}{\pi\ell^2 \cosh^2(\xi\Omega_\omega/\ell)} \quad (30)$$

where we have introduced a coefficient ξ related to the fiber parameters and to the second-order PMD coefficient as follows:

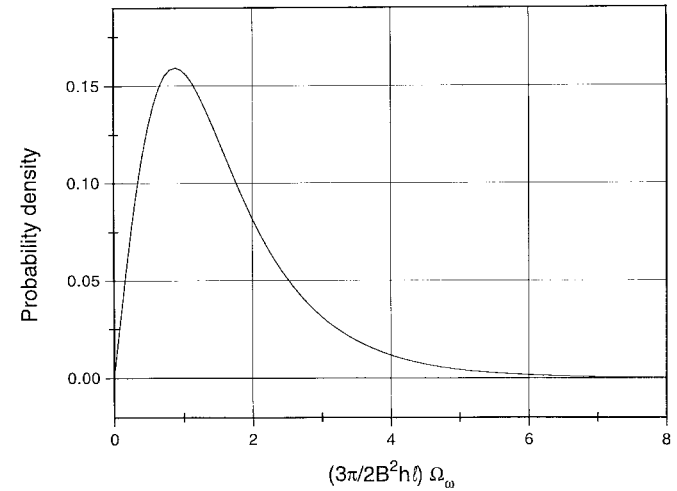
$$\xi \equiv \frac{3\pi}{2B^2h} = \frac{\sqrt{3}\pi u}{2\text{PMD}_{\text{coeff}}^{\text{second-order}}}. \quad (31)$$

This distribution function is illustrated in Fig. 3 (linear scale) and Fig. 4 (log scale). Fig. 5 represents the probability distribution $\tilde{\rho}_\ell$ for the distortion parameter $\tilde{\eta}^{(2)}$ (in case the chromatic dispersion is negligible) $\tilde{\rho}_\ell[\tilde{\eta}^{(2)}] = [6/(\gamma\omega_m\sqrt{3\tilde{\eta}^{(2)}})]\rho_\ell[(4/\gamma\omega_m)\sqrt{3\tilde{\eta}^{(2)}}]$. Accordingly, the probabilities that the actual values of the second-order PMD Ω_ω and of the distortion parameter $\tilde{\eta}^{(2)}$ are larger than n times their rms and mean values, respectively, are given in Table II where $\sqrt{\langle\tilde{\Omega}_\omega^2\rangle} = (\ell/u)\text{PMD}_{\text{coeff}}^{\text{second-order}}$ and

$$\text{Prob}\left(\Omega_\omega > n \cdot \sqrt{\langle\tilde{\Omega}_\omega^2\rangle}\right) = \int_{n\sqrt{\langle\tilde{\Omega}_\omega^2\rangle}}^{+\infty} \rho_\ell(\Omega_\omega) d\Omega_\omega. \quad (32)$$

For the distortion parameter $\tilde{\eta}^{(2)}$ one has

$$\begin{aligned} \text{Prob}(\tilde{\eta}^{(2)} \geq n\langle\tilde{\eta}^{(2)}\rangle) &= \text{Prob}(\tilde{\Omega}_\omega^2 \geq n\langle\tilde{\Omega}_\omega^2\rangle) \\ &= \text{Prob}(\Omega_\omega \geq \sqrt{n}\sqrt{\langle\tilde{\Omega}_\omega^2\rangle}). \end{aligned} \quad (33)$$

Fig. 5. Probability density distribution of the distortion parameter $\eta^{(2)}$.

V. EXPERIMENTAL RESULTS

There are several possible tests of the present theory. The first one tests the relation (27) between the first- and second-order PMD coefficients. The second one tests the relation (6) between the second-order PMD coefficient and the distortion parameter $\eta^{(2)}$. Results concerning both kinds of experimental tests are presented below, whereas numerical simulation results are presented in the next section.

First, in order to test experimentally the relation (27) between the first and second-order PMD coefficients, we have used a polarimeter (HP 8509B) with a noncommercial acquisition software and homemade analysis. The acquisition software provide the Jones matrix elements in function of wavelength. From these, the principal state vectors $\vec{\Omega}(\omega)$ and their derivative $\vec{\Omega}_\omega(\omega)$ can be computed. The wavelength average of the derivative are then compared to the first-order PMD (measured using the interferometric technique [9]). Table III presents results obtained on a cabled fiber and on a concatenation of HiBi fibers, both around 1300 nm and at 1550 nm. The results are in good agreement with the theory.

TABLE II
PROBABILITY THAT THE ACTUAL SECOND-ORDER PMD Ω_ω AND THE DISTORTION
PARAMETER $\bar{\eta}^{(2)}$ ARE n TIMES LARGER THAN THEIR MEAN VALUES

n	$Prob(\Omega_\omega > n \cdot \sqrt{\langle \bar{\Omega}_\omega^2 \rangle})$	$Prob(\bar{\eta}^{(2)} > n \cdot \langle \bar{\eta}^{(2)} \rangle)$
0	1	1
1	$3.11 \cdot 10^{-1}$	$3.11 \cdot 10^{-1}$
2	$3.55 \cdot 10^{-2}$	$1.31 \cdot 10^{-1}$
3	$3.33 \cdot 10^{-3}$	$6.53 \cdot 10^{-2}$
4	$2.84 \cdot 10^{-4}$	$3.55 \cdot 10^{-2}$
5	$2.30 \cdot 10^{-5}$	$2.06 \cdot 10^{-2}$
6	$1.79 \cdot 10^{-6}$	$1.24 \cdot 10^{-2}$
7	$1.37 \cdot 10^{-7}$	$7.80 \cdot 10^{-3}$
8	$1.02 \cdot 10^{-8}$	$5.03 \cdot 10^{-3}$
9	$7.53 \cdot 10^{-10}$	$3.33 \cdot 10^{-3}$
10	$5.49 \cdot 10^{-11}$	$2.24 \cdot 10^{-3}$

TABLE III
EXPERIMENTAL COMPARISON OF FIRST- AND SECOND-ORDER PMD. ACCORDING
TO RELATION (27) THE TWO LAST COLUMNS SHOULD HAVE IDENTICAL
VALUES. TAKING INTO ACCOUNT THE FINITE STATISTICAL SAMPLE (SEE
SECTION VI), THESE RESULTS ARE IN GOOD AGREEMENT WITH THE THEORY

fiber	Wavelength range [nm]	1st order PMD delay $\Delta\tau$ [ps]	2nd order PMD $\sqrt{\langle \bar{\Omega}_\omega^2 \rangle} > \omega$ [ps ²]	$\frac{\Delta\tau^2}{\sqrt{3}}$
Cabled fiber	1260-1330	0.585	0.23	0.20
Cabled fiber	1500-1560	0.675	0.33	0.26
Concatenation HiBi	1260-1330	5.83	20.00	19.63
Concatenation HiBi	1520-1570	5.31	21.41	16.31

Next, in order to experimentally test the relations (6) and (12), we measured the distortion products in an analog SCM transmission system, specifically its CSO distortion, and tried to relate it to the PMD of the fiber link. This is a nontrivial test, since many other causes may contribute to the distortion. To measure the CSO distortion we worked with the signal from a CATV-head-end delivering 34 synchronously modulated video carriers. These were used to modulate a 1310-nm DFB-laser from a commercial CATV transmitter-receiver system. The RF-signal was launched directly to the CATV transmitter and, after going through the fiber under test, was sent to the receiver. From there on, the signal entered the spectrum analyzer to measure the corresponding CSO distortion. Several samples of cabled and uncabled fibers with different PMD values were used for these measurements. To take into account the system contribution to CSO, a short fiber connection was used to get a reference distortion. Accordingly, the desired CSO parameters were obtained by subtracting the measured distortion with the short optical fiber from the one measured with the test fibers. Measurements were performed during long periods of time to average out the fluctuations of CSO. The results are presented in Fig. 6. CSO is given as the carrier to distortion ratio in dB, dBc. There is a clear systematic difference between the theoretical predictions, based on (6) and (12), and the experimental data, despite our effort to reconcile them. We believe that an unmastered source of distortion affects these data. Though it should be noted that the theoretical prediction are within two sigma from the experimental data.

An other comparison of the theory with experiments is possible through comparing the statistical properties of the

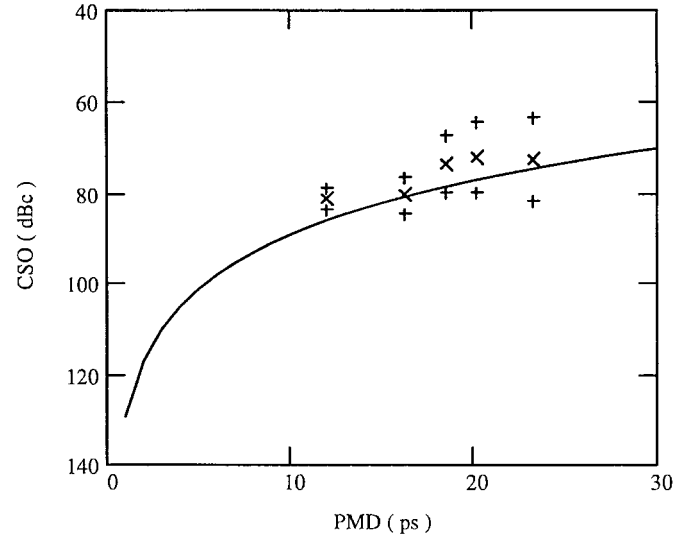


Fig. 6. Comparison between theory (line) and experiment (cross) for 34 channels CSO versus PMD. The (+) symbols indicate an error range of ± 2 sigma. The theoretical curves have been calculated with formula (12) and with the following parameters: $w = 744.5$ MHz, $m = 5\%$, $h = 150$ MHz/mA, $I_B - I_{TH} = 50$ mA, $N = 8$. (m , h , and I_x are taken from the specifications of the transmission system).

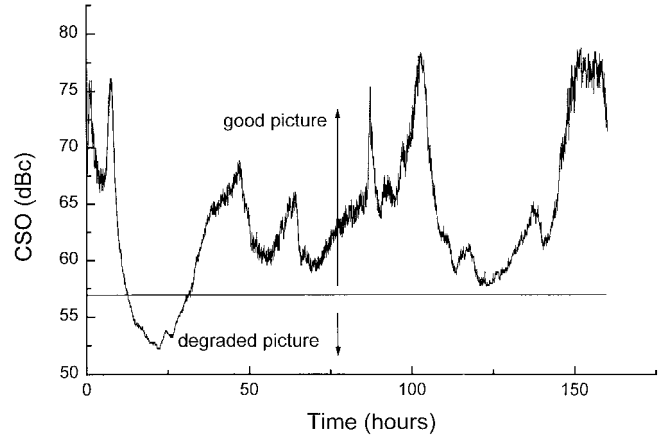


Fig. 7. Variation of CSO over time measured in an installed fiber with 29 ps of PMD. The transmitter was modulated with two sinusoidal signals. The full-time scale corresponds to one week time.

time dependence of CSO. Indeed, one expects that all the fluctuations are produced by the PMD causes. With fibers of low PMD, the variation over time was confirmed to be much lower. Hence, analyzing these fluctuations should provide direct information about the PMD cause, not influenced by other static causes. For this purpose, an installed fiber, buried in the ground, showing a high PMD delay of 29 ps, was used as a sample for measuring CSO. Fig. 7 shows the results for an entire week. Here, instead of the 34 modulated carriers, we used only two unmodulated RF-channels, and CSO was evaluated at the sum of the two frequencies. Clearly the CSO undergoes large fluctuations. These fluctuations are analyzed in Fig. 8, which shows the theoretical and measured probabilities for CSO to stay below a certain threshold. The theoretical values are taken from the second column of Table II. Now, the agreement between theory and experimental data is striking. Hence, our confidence that the theory is reliable. This confi-

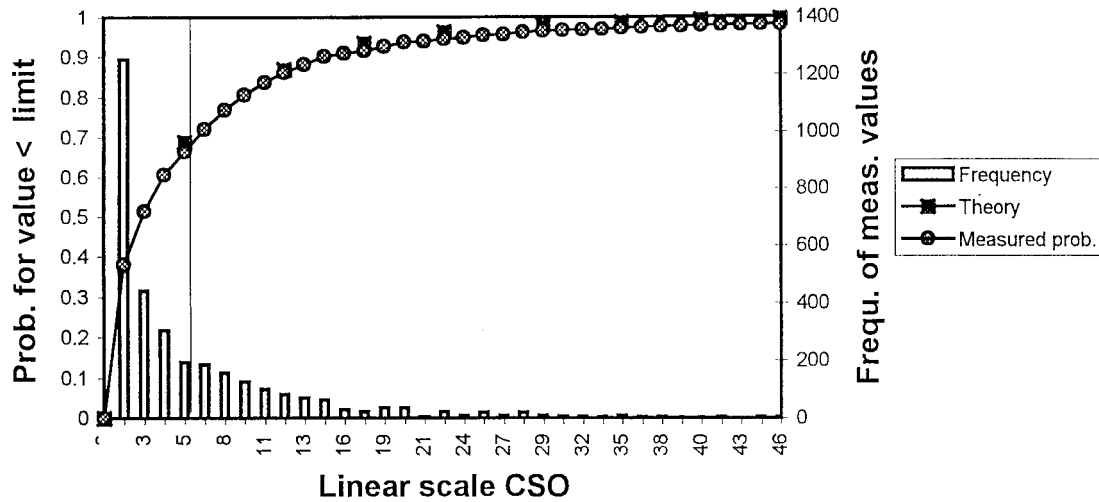


Fig. 8. Probability of CSO to stay below a certain value (left side scale), for linearly scaled CSO ($= 10^{-\text{dB}/10} \cdot 10^7$, i.e., rightmost point corresponds to the highest CSO level, 54 dBc). Comparison between theory (squares) and experiment (circles). The vertical line indicates the position of the average linear value (62 dBc). The histogram bars show the measured frequency of occurrence of the CSO values (right side scale). Measurements were done with a fiber of 18.5 ps PMD delay and with two RF-channels. The temperature of the sample was cycled in a climatic chamber.

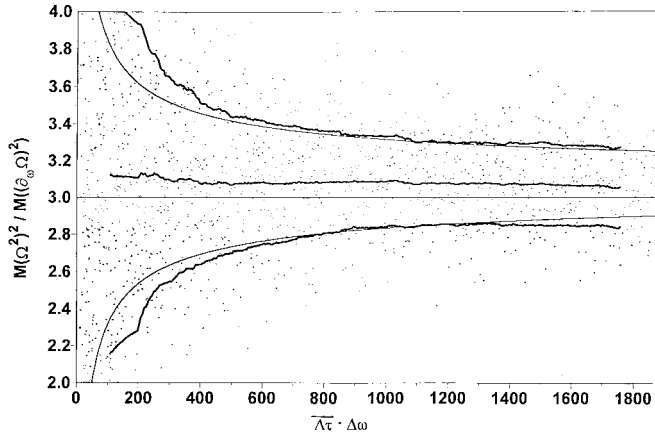


Fig. 9. Numerical test of the predicted relation (27) between the first and second-order PMD. The fiber is simulated as a concatenation of 200 fiber trunks of random birefringence and orientation. All the polarization mode coupling takes place at the junctions between the trunks. The modal birefringence of each trunk is modeled as independent of wavelength, so that large wavelength ranges can be considered. The theoretical mean PMD delay $\Delta\tau$ was computed as the RMS mean taken over a very large frequency ranges of 50 000 [1/ps] [19]. First- and second-order PMD's were computed over different wavelength ranges and around different central wavelengths, providing the 1000 ratios displayed. The mean of these ratios is indicated (central a bold line), it is close to the predicted ratio of three. The upper and lowest bold lines represent the mean value plus and minus one standard deviation, illustrating the large fluctuations due to the statistical nature of PMD. The standard deviation was fitted by $\text{cste}/\sqrt{\Delta\tau \Delta\omega}$ curve and the value 7.6 was found for the constant.

dence is reinforced by the observation that the fit of the data on Fig. 8 assuming a Maxwellian distribution for the second-order PMD instead of the distribution (30) is significantly worse.

VI. MEASUREMENT ACCURACY LIMITATION DUE TO THE STATISTICAL NATURE OF PMD

In order to test further the predicted relation between first- and second-order PMD and to study the accuracy limitation imposed on measurements of second-order PMD by the statistical nature of the phenomenon, we have used our software

simulation to compute both of them over the same wavelength range. This software models optical fibers as concatenations of many birefringent fiber trunks without polarization mode coupling, all the coupling taking place at the junctions between these trunks. For each run, 200 fiber trunks of random birefringence and random orientation are chosen, using random number generators. The modal birefringences are assumed to be wavelength independent. Then the first-order PMD is computed as the RMS value of the DGD (Differential Group Delays) over a very large optical frequency range of 50 000 1/ps. Next, first and second-order PMD values are computed for many different frequency ranges and each ratio is plotted, see Fig. 9. Note that different ranges may be of the same length, like, for example, from 1200–1250 nm and from 1300 to 1350 nm. Hence, many ratios of first- to second-order PMD values can be computed on the same fiber (a thousand are displayed on Fig. 9). The horizontal scale is the product of this range (expressed in optical frequencies with 1/ps units) times the theoretical mean first-order PMD delay (expressed in ps). Each of the 1500 data points represents one computed ratio of the mean values. These results show also that whenever the product between the first-order PMD delay and the optical frequency range is small, the fluctuations around the above ratio are large. Recall that a wavelength range of 100 nm corresponds to about $\Delta\omega = 100(1/\text{ps})$. However, when either the PMD delay or the optical frequency range is large enough, the fluctuation decrease, as expected. The results show also a small systematic difference with respect to the predicted ratio of three. This difference seems to be due to the number 200 of trunks we used for this simulation. Indeed, using fewer trunks increases this small discrepancy. But since increasing the number of trunks increases significantly the computation time, we limited our simulation to the 200 trunk case. One data point, however, was computed for 800 trunks and a product $\Delta\tau \Delta\omega = 50000$. For such a large value the fluctuation should be negligible, the result was 3.03, acceptably close to the predicted value of 3.00. Fig. 9 presents also the standard

deviation, see the bold upper and lower curves. Although we do not have a theoretical formula to fit to this standard deviation, we fitted a function $\chi/\sqrt{\Delta\tau\Delta\omega}$. This fit is displayed with the value 7.6 for the constant χ .

VII. IMPLICATION FOR STANDARDIZATION

In this section we suggest some orders of magnitude for future specifications. The idea is to base these specifications on the existing ones for chromatic dispersion. Indeed, since second-order PMD has the same effect on analog systems as chromatic dispersion, it is natural to incorporate their effect in the existing ITU-T specification [10]

$$\frac{1}{\ell} \left[\tau' - \frac{\vec{\Omega}_\omega(\omega_0) \vec{m}_{\text{laser}}}{2} \right] \leq 3.5 \quad \left[\frac{\text{ps}}{\text{nm km}} \right] \quad (34)$$

where the 3.5 ps/(nm.km) is the present day specification for chromatic dispersion at 1310 nm. If one desires to maintain this upper limit, one has to choose how much one may allocate to chromatic dispersion and how much may be due to second-order PMD. A sensible choice is

$$\frac{\tau'}{\ell} \leq 3 \quad \left[\frac{\text{ps}}{\text{nm km}} \right] \quad (35)$$

where as in Section II, $\tau' = d\tau/d\omega$. Hence

$$\text{PMD}_{\text{coeff}}^{\text{second-order}} \leq \frac{1}{5} \frac{u}{\ell} \sqrt{\langle \vec{\Omega}_\omega^2 \rangle} \leq 0.2 \left[\frac{\text{ps}}{\text{nm km}} \right] \quad (36)$$

where the factor of five takes into account the large fluctuations that affect second-order PMD (see Section IV-B). Now, using the relation (29) between first and second-order PMD, the above limit leads to

$$\text{PMD}_{\text{coeff}}^{\text{first-order}} \leq \sqrt{0.2\sqrt{3}} \approx 0.5 \left[\frac{\text{ps}}{\sqrt{\text{km}}} \right]. \quad (37)$$

This value of 0.5 [ps/√km] matches the proposed specification presently under discussion at all major standardization bodies.

VIII. EFFECTS OF POLARIZATION DEPENDENT LOSSES (PDL)

In this section we summarize some preliminary results about the combined effects of PDL and PMD on analog transmission systems. Further results can be found in [11] and [12].

A. Case of Polarization Sensitive Detectors

This is the case studied by Poole in [1]. It assumes that the efficiency of the detection scheme is sensitive to polarization, that is that some polarization states are detected with a higher efficiency than others. Note that this differs from PDL inside or along the transmission line as discussed briefly in the next section.

Let $\mathcal{T}$ denote the efficiency of the detection scheme, i.e., the transmission coefficient between the light signal emerging from the fiber to the electrical signal out of the detector. We assume that $\mathcal{T}$ may depend on the polarization state $\vec{M}_{\text{out}}$ of the light signal. Using the notations of [11] and [12], one has

$$\mathcal{T} = \mathcal{T}_{\text{depol}} (1 + \vec{M}_{\text{out}} \vec{\Gamma}) \quad (38)$$

where $\mathcal{T}_{\text{depol}}$ is the detection efficiency for unpolarized light and $\vec{\Gamma}$ is the PDL vector [11]. $\vec{\Gamma}$ is parallel to the less attenuated polarization state on the Poincaré sphere and is of norm $\Gamma = (\mathcal{T}_{\text{max}} - \mathcal{T}_{\text{min}})/(\mathcal{T}_{\text{max}} + \mathcal{T}_{\text{min}})$. Because of the birefringence of the fiber, the polarization state of the light signal that impacts on the detector depends on the optical frequency ω

$$\frac{d\vec{M}_{\text{out}}}{d\omega} = \vec{\Omega} \wedge \vec{M}_{\text{out}} \quad (39)$$

where $\vec{\Omega}$ is the principal polarization vector. Accordingly, $\mathcal{T}$ depends also on the optical frequency

$$\mathcal{T}(\omega_0 + \delta\omega) = \mathcal{T}(\omega_0) + \mathcal{T}_{\text{depol}} \vec{\Gamma} (\vec{\Omega} \wedge \vec{M}_{\text{out}}) \delta\omega. \quad (40)$$

In the case of a purely sinusoidal signal $\delta\omega = \gamma \sin(\omega_m t)$, see (2) and the contribution of a polarization sensitive detector to harmonic generation is given by

$$\eta_{\text{PDL}}^{(2)} = \frac{\gamma^2}{4} [\vec{\Gamma} (\vec{\Omega} \wedge \vec{M}_{\text{out}})]^2. \quad (41)$$

This result agrees with Poole's [1] (his ΔT is defined as one-half of our Γ).

B. Case of Polarization Dependent Losses in a Birefringent Optical Link

The general problem of the combined effects of PDL and PMD is quite more complex than the case studied in the previous subsection. Indeed, whenever the elements with PDL are “sandwiched” between fibers with some PMD, the principal polarization states are no longer mutually orthogonal. Consequently, the outcoming intensity contains an interference term between these two principal states. This interference term can reduce or increase the spread of optical pulses and can induce additional distortions in analog transmissions. Another consequence of the combination of PDL and PMD is that the attenuation for unpolarized light and the global PDL are wavelength dependent, similarly to the wavelength dependence of the DGD. Hence, they require a statistical description and characterization. A general description of such complex phenomena is beyond the present article and will be developed elsewhere [12]. First experimental results can be found in [13]

IX. CONCLUSION

The effects of second-order PMD are similar to the ones due to chromatic dispersion, except that their effects fluctuate with time like all polarization effects in standard single-mode fibers. For analog systems, we have derived a relation relating the distortion parameters $\eta^{(2)}$ and CSO to the principal polarization state vector, in agreement with [1]. From this, a definition of second-order PMD has been proposed. For digital systems, pulse spreadings have been studied and the same quantity, defined as second-order PMD, appears as the dominating term. The statistics of second-order PMD has been studied theoretically. In particular its fiber length dependence and its relation to first-order PMD have been predicted. These predictions have been tested by numerical simulations and

experimental tests. Finally, some orders of magnitude for future specifications have been discussed.

APPENDIX A DISTORTION DUE TO SECOND-ORDER PMD

In this Appendix, we compute the distortion due to second-order PMD. We will work in the time domain, contrary to [1] where the same formula was derived first in the frequency domain. For simplicity, we consider only a pure sinus signal, but the formula generalizes to an arbitrary number of signals modulated at different frequencies.

Consider a sinusoidal signal I_{in} of modulation frequency ω_m launched into the fiber

$$I_{\text{in}}(t) = I_0[1 + \alpha \sin(\omega_m t)]. \quad (42)$$

The instantaneous optical laser frequency $\omega(t)$ is directly proportional to the laser output power I_{in} $\omega(t) = \omega_0 + \gamma \sin(\omega_m t)$ where γ is related to the modulation amplitude, the laser chirp coefficient and the laser gain. More explicitly one has $\gamma = (\eta \delta I)/2$ with η is the laser chirp in [GHz/mA] and δI the peak to peak current modulation (in mA).

Let $\pm \vec{\Omega}(\omega)$ denote the two input principal states of polarization. At the end of the fiber, the power associated with the slow mode $-\vec{\Omega}$ is delayed with respect to the fast mode $+\vec{\Omega}$ by a time $\delta\tau(\omega)$. Distortion are produced by two different causes. First, the propagation constants of each polarization mode is modulated because of the laser's chirp. This effect is of the same nature as chromatic dispersion, but the frequency dependence of the DGD, $\delta\tau(\omega)$ adds to chromatic dispersion. Denoting $\tau(\omega) \pm \frac{1}{2}\delta\tau(\omega)$ the time of propagation through the fiber, this effect produces the following intensity at the fiber output:

$$I_{\text{out,fast/slow}}(t) = I_0 \{ 1 + \alpha \sin(\omega_m t) - 2\omega_m \alpha \cos(\omega_m t) \cdot [\tau(\omega_0) \mp \frac{1}{2}\delta\tau(\omega_0)] - \omega_m \alpha \gamma \sin(2\omega_m t) \cdot [\tau'(\omega_0) \mp \frac{1}{2}\delta\tau'(\omega_0)] \} + o(\gamma^2) \quad (43)$$

where the coefficient of the term in γ corresponds to the well known distortion due to chromatic dispersion [14], [15], and $\tau'(\omega_0) = (d\tau/d\omega)_{\omega=\omega_0}$ and $\delta\tau'(\omega_0) = (d\delta\tau/d\omega)_{\omega=\omega_0}$.

Next, a second cause of distortion is the frequency dependence of the principal polarization state. Consequently, the coupling between the laser, assumed to emit light of fixed polarization $\vec{m}_{\text{laser}}$, and each of the two polarization modes is modulated. Indeed the coupling factors are given by Malus Law: $\{1 \pm \hat{\Omega}[\omega(t)]\vec{m}_{\text{laser}}/2\}$ where $\hat{\Omega}[\omega(t)] = \{\vec{\Omega}[\omega(t)]/\delta\tau[\omega(t)]\}$ is the unit vector in the direction $\vec{\Omega}(\omega(t))$. In order to evaluate this effect, we expand $\vec{\Omega}$ and $\hat{\Omega}$ in γ up to first order

$$\vec{\Omega}[\omega(t)] = \vec{\Omega}(\omega_0) + \vec{\Omega}_\omega(\omega_0)\gamma \sin(\omega_m t) + o(\gamma^2) \quad (44)$$

where

$$\vec{\Omega}_\omega(\omega_0) = \left(\frac{d\vec{\Omega}}{d\omega} \right)_{\omega=\omega_0} \quad (45)$$

and

$$\hat{\Omega}[\omega(t)] = \frac{\vec{\Omega}(\omega_0)}{\delta\tau(\omega_0)} + \gamma \frac{\sin(\omega_m t)\vec{\Omega}_\omega(\omega_0)}{\delta\tau(\omega_0)} - \gamma \frac{\sin(\omega_m t)\vec{\Omega}(\omega_0)\delta\tau'(\omega_0)}{\delta\tau^2(\omega_0)} + o(\gamma^2). \quad (46)$$

Finally, one obtains the output signal

$$I_{\text{out}}(t) = I_{\text{out,fast}}(t) \frac{1 + \hat{\Omega}[\omega(t)]\vec{m}_{\text{laser}}}{2} + I_{\text{out,slow}}(t) \frac{1 - \hat{\Omega}[\omega(t)]\vec{m}_{\text{laser}}}{2} \quad (47)$$

$$= I_0 \left\{ 1 + \alpha \sin(\omega_m t) - 2\alpha\omega_m \cos(\omega_m t) [\tau(\omega_0) + \frac{1}{2}\vec{\Omega}(\omega_0)\vec{m}_{\text{laser}}] - \alpha\gamma\omega_m \sin(2\omega_m t) [\tau(\omega_0) - \frac{1}{2}\vec{\Omega}_\omega(\omega_0)\vec{m}_{\text{laser}}] \right\} + o(\gamma^2). \quad (48)$$

The first harmonic $\eta^{(2)}$ is thus given by [1]

$$\eta^{(2)} = \frac{(\text{RFat } 2\omega_m)_{\text{out}}}{(\text{RFat } \omega_m)_{\text{in}}} = (\gamma\omega_m)^2 [\tau'(\omega_0) - \frac{1}{2}\vec{\Omega}_\omega(\omega_0)\vec{m}_{\text{laser}}]^2. \quad (49)$$

This is (4) of the main text. Accordingly, to first order in γ , the relevant statistical quantity is $\vec{\Omega}_\omega$. Note that we have implicitly assumed that the two principal polarization states are mutually orthogonal. Indeed, if this would not be the case, then one could not simply add their intensities, one would have to take into account interference terms. Consequently, in the presence of PDL, when the principal states are generally not orthogonal, the analyzes is quite more involved and one may expect further effects producing a significantly larger distortion.

APPENDIX B PULSE SPREADING DUE TO CHROMATIC DISPERSION AND FIRST- AND SECOND-ORDER PMD

In this Appendix the output pulse spread σ_{out} is computed, followed by the proof that the rate of rotation of the input and output principal polarization states are equal: $\alpha_{\text{in}} = \alpha_{\text{out}}$.

A. Spread of Output Pulse

Dropping the higher order terms in the formulas (16)–(19) and using the following notations $a_+ = \langle \psi_{\text{in}0}^+ | \psi_{\text{in}} \rangle$ and $a_- = \langle \psi_{\text{in}0}^- | \psi_{\text{in}} \rangle$, we obtain (50) shown at the bottom of the next page where some irrelevant phases have been absorbed into the normalization constant N and, for simplicity, we moved the origin of time such that $t_0 = 0$. Next, we use the two following Taylor expansions:

$$e^{-i\delta\omega^2(\tau' \pm \delta\tau'_0)/2} = 1 - i\delta\omega^2 \frac{(\tau' \pm \delta\tau'_0)}{2} - \delta\omega^4 \frac{(\tau' \pm \delta\tau'_0)^2}{8} + O[\delta\omega^6 \cdot (\tau' \pm \delta\tau'_0)^3] \quad (51)$$

$$1/(1 + \delta\omega^2 \cdot \alpha^2) = 1 - \delta\omega^2 \cdot \alpha^2 + \delta\omega^4 \cdot \alpha^4 + O(\delta\omega^6 \cdot \alpha^6). \quad (52)$$

The inverse Fourier transform can now be computed, bringing us back to the time domain. The spread of the output pulse is defined as

$$\sigma_{\text{out}}^2 \equiv \frac{\int t^2 |E_{\text{out}}(t)|^2 dt}{\int |E_{\text{out}}(t)|^2 dt} - \left\{ \frac{\int t |E_{\text{out}}(t)|^2 dt}{\int |E_{\text{out}}(t)|^2 dt} \right\}^2. \quad (53)$$

This spread depends on the input polarization state $\psi_{\text{in}} = a_+ \psi_0^+ + a_- \psi_0^-$. Rearranging the terms one obtains (54) shown at the bottom of the page where $\Omega_\omega^2 \equiv (\vec{\Omega}/d\omega)^2 = 4\alpha^2 \cdot \delta\tau^2 + (\delta\tau')^2$ is the second-order PMD as defined in the first section.

Using the notation $a_+ = \sqrt{(1+\eta)/2} \cdot e^{-i\varphi/2}$ and $a_- = \sqrt{(1-\eta)/2} \cdot e^{i\varphi/2}$, the corresponding polarization on the Poincaré sphere reads $\vec{m} = (m_1, m_2, m_3) = [\sqrt{1-\eta^2} \cdot \cos(\varphi), \sqrt{1-\eta^2} \cdot \sin(\varphi), \eta]$. Hence, one recovers the results presented in Table I of Section II

$$\begin{aligned} \sigma_{\text{out}}^2 &= \sigma_{\text{in}}^2 + \frac{1}{4}\delta\tau^2 \cdot (1 - m_z^2) + \frac{1}{4}\tau'^2 \cdot \Delta\omega^2 \\ &+ \left(\frac{1}{4}\Omega_\omega^2 + \alpha \cdot \tau' \cdot \delta\tau \cdot m_1 + \frac{1}{2}\tau' \cdot \delta\tau' \cdot m_z \right) \cdot \Delta\omega^2 \\ &+ \left[-\frac{3}{4}\alpha^2 \cdot \delta\tau^2 \cdot (1 - m_z^2) - \frac{3}{4}\alpha \cdot \delta\tau \cdot \delta\tau' \cdot m \cdot m_z \right] \\ &\cdot \Delta\omega^2 + \frac{1}{8}\alpha \cdot \delta\tau^3 \cdot m_2 \cdot m_z \cdot \Delta\omega^2. \end{aligned} \quad (55)$$

Proof that $\alpha_{\text{in}} = \alpha_{\text{out}}$: We shall need the following equations taken from [7]: $\vec{\Omega}_{\text{in}}\vec{\sigma} = \mp i(U_\omega^\dagger \partial_\omega)U_\omega = \pm i(\partial_\omega U_\omega^\dagger)U_\omega$ and $\vec{\Omega}_{\text{out}}\vec{\sigma} = U_\omega \vec{\Omega}_{\text{in}} \vec{\sigma} U_\omega^\dagger$ where U_ω is the unitary complex 2×2 matrix describing the change of polarization of a light signal at frequency ω when propagating through the fiber, and $\vec{\sigma}$ represents a vector containing the three Pauli matrices, and $\dagger$ stands for the adjoint matrix. Using the notation $\delta\tau(\omega) = \|\vec{\Omega}_{\text{in}}\| = \|\vec{\Omega}_{\text{out}}\|$, $\hat{\Omega}_{\text{in}} = \vec{\Omega}_{\text{in}}/\delta\tau$, $\hat{\Omega}_{\text{out}} = \vec{\Omega}_{\text{out}}/\delta\tau$, we compute:

$$\begin{aligned} U_\omega^\dagger (\partial_\omega \hat{\Omega}_{\text{out}} \vec{\sigma}) U_\omega &= U_\omega^\dagger (\partial_\omega U_\omega) \hat{\Omega}_{\text{in}} \vec{\sigma} + \partial_\omega \hat{\Omega}_{\text{in}} \vec{\sigma} \\ &+ \hat{\Omega}_{\text{in}} \vec{\sigma} (\partial_\omega U_\omega^\dagger) U_\omega \\ &= \partial_\omega \hat{\Omega}_{\text{in}} \vec{\sigma} \end{aligned} \quad (56)$$

where we have used the following relation among the Pauli matrices: $\vec{\Omega}\vec{\sigma} \cdot \hat{\Omega}\vec{\sigma} = \vec{\Omega}\hat{\Omega} = \delta\tau$. Taking the trace of the square

of both sides of (56) one obtains:

$$\begin{aligned} 4\alpha_{\text{out}}^2 &= \|\partial_\omega \hat{\Omega}_{\text{out}}\|^2 = \text{Tr}(\partial_\omega \hat{\Omega}_{\text{out}} \vec{\sigma})^2 = \text{Tr}(\partial_\omega \hat{\Omega}_{\text{in}} \vec{\sigma})^2 \\ &= \|\partial_\omega \hat{\Omega}_{\text{in}}\|^2 = 4\alpha_{\text{in}}^2, \end{aligned} \quad (57)$$

which proves that $\alpha_{\text{in}} = \alpha_{\text{out}}$.

APPENDIX C

DERIVATION OF THE SECOND-ORDER PMD LENGTH DEPENDENCE

To compute the second-order PMD length dependence, we start from the dynamical equation for first-order PMD and derive the dynamical equation for second-order PMD. Then, we show that the statistics of second-order PMD is independent of the optical frequency ω . Finally, we use stochastic calculus to solve the dynamical equation for second-order PMD in the case $\omega = 0$.

A. Dynamical Equation for Second-Order PMD

The dynamical equation for first-order PMD reads

$$\frac{d\vec{\Omega}(z, \omega)}{dz} = \vec{\beta}(z) + \omega \vec{\beta}(z) \wedge \vec{\Omega}(z, \omega). \quad (58)$$

Denoting $\vec{\Omega}_\omega = (d\vec{\Omega}/d\omega)$ the derivative of $\vec{\Omega}$ with respect to the optical frequency ω , one easily derives the following dynamical equation for second-order PMD by differentiation with respect to ω :

$$\frac{d\vec{\Omega}_\omega}{dz} = \vec{\beta}(z) \wedge \vec{\Omega}_\omega(z, \omega) + \omega \vec{\beta}(z) \wedge \vec{\Omega}_\omega(z, \omega) \quad (59)$$

where $\vec{\beta}(z)$ represents the local modal birefringence in [ps/km] which fluctuates along the fiber, with zero mean and correlation length h [3]

$$\langle \vec{\beta}(z) \rangle = \vec{0} \quad \text{and} \quad \langle \vec{\beta}(z) \vec{\beta}(z') \rangle = B^2 e^{-(2|z-z'|)/h}. \quad (60)$$

$\vec{\beta}$ can be generated by the following stochastic equation:

$$d\vec{\beta} = -\frac{2}{h} \vec{\beta} dz + \frac{2B}{\sqrt{3h}} d\vec{W}. \quad (61)$$

$$\begin{aligned} E_{\text{out}}(\omega) &= N e^{-\delta\omega^2/\Delta\omega^2} \cdot \left(e^{i[-\delta\omega\delta\tau_0/2 - \delta\omega^2(\tau' + \delta\tau'_0)/2]} \cdot (a_+ + \delta\omega \cdot \alpha \cdot a_-)/(1 + \delta\omega^2\alpha^2) \cdot \psi_0^+ \right. \\ &+ e^{i[\delta\omega\delta\tau_0/2 - \delta\omega^2(\tau' - \delta\tau'_0)/2]} \cdot (-\delta\omega \cdot \alpha \cdot a_- + \delta\omega^2\alpha^2 a_+)/ (1 + \delta\omega^2\alpha^2) \cdot \psi_0^+ \\ &+ e^{i[-\delta\omega\delta\tau_0/2 - \delta\omega^2(\tau' + \delta\tau'_0)/2]} \cdot (\delta\omega \cdot \alpha \cdot a_+ + \delta\omega^2\alpha^2 a_-)/ (1 + \delta\omega^2\alpha^2) \cdot \psi_0^- \\ &\left. + e^{i[\delta\omega\delta\tau_0/2 - \delta\omega^2(\tau' - \delta\tau'_0)/2]} \cdot (a_- - \delta\omega \cdot \alpha \cdot a_+)/ (1 + \delta\omega^2\alpha^2) \cdot \psi_0^- \right) \end{aligned} \quad (50)$$

$$\begin{aligned} \sigma_{\text{out}}^2 &= \sigma_{\text{in}}^2 + \delta\tau^2 \cdot |a_+|^2 \cdot |a_-|^2 + \frac{1}{4}\tau'^2 \cdot \Delta\omega^2 \\ &+ \left[\frac{1}{4}\Omega_\omega^2 + 2\alpha \cdot \tau' \cdot \delta\tau \cdot \text{Re}(a_+ \cdot a_-^*) + \frac{1}{2}\tau' \cdot \delta\tau' \cdot (|a_+|^2 - |a_-|^2) \right] \cdot \Delta\omega^2 \\ &+ \left[-3\alpha^2 \cdot \delta\tau^2 \cdot |a_+|^2 \cdot |a_-|^2 - \frac{3}{2}\alpha \cdot \delta\tau \cdot \delta\tau' \cdot \text{Re}(a_+ \cdot a_-^*) \cdot (|a_+|^2 - |a_-|^2) \right] \cdot \Delta\omega^2 \\ &+ \frac{1}{4}\alpha \cdot \delta\tau^3 \cdot \text{Im}(a_+^* \cdot a_-) \cdot (|a_+|^2 - |a_-|^2) \cdot \Delta\omega^2 \end{aligned} \quad (54)$$

Where for $i = 1, 2, 3$, dW_i are three independent Wiener processes

$$\forall i, \langle dW_i \rangle = 0 \quad (62)$$

$$\forall i, j, \langle dW_i dW_j \rangle = \delta_{ij} dz. \quad (63)$$

The birefringence B and the correlation length h are chosen such that $\langle \vec{\Omega}^2 \rangle$ agrees with the known result for first-order PMD [3]

$$\langle \vec{\Omega}^2 \rangle = \frac{B^2 h^2}{2} \left(2 \frac{z}{h} - 1 + e^{-(2z/h)} \right). \quad (64)$$

At the entrance of the fiber, the initial conditions are

$$\vec{\Omega}(z=0, \omega) = \vec{0} \quad \vec{\Omega}_\omega(z=0, \omega) = \vec{0}. \quad (65)$$

B. Closed Set of Equation for $\vec{\Omega}_\omega^2$

We are looking for a set of equations determining $\langle \vec{\Omega}_\omega^2 \rangle$ and such that if we integrate the last one, going reversely we can integrate all equations. To evaluate the different means, we rewrite (59) in the form

$$d\vec{\Omega}_\omega = \vec{\beta}(z) \wedge \vec{\Omega}(z, \omega) dz + \omega \vec{\beta}(z) \wedge \vec{\Omega}_\omega(z, \omega) dz. \quad (66)$$

Hence, we find

$$d\vec{\Omega}_\omega^2 = 2\vec{\Omega}_\omega d\vec{\Omega}_\omega = 2\vec{\Omega}_\omega(\vec{\beta} \wedge \vec{\Omega}) dz. \quad (67)$$

To compute the mean of $\vec{\Omega}_\omega^2$, we need the mean of $\vec{\Omega}_\omega(\vec{\beta} \wedge \vec{\Omega})$

$$d\vec{\Omega}_\omega(\vec{\beta} \wedge \vec{\Omega}) = -\frac{2}{h} \vec{\Omega}_\omega(\vec{\beta} \wedge \vec{\Omega}) dz + D\vec{\Omega}_\omega(d\vec{W} \wedge \vec{\Omega}) + (\vec{\beta} \wedge \vec{\Omega})^2 dz. \quad (68)$$

Now, we need to compute $(\vec{\beta} \wedge \vec{\Omega})^2$

$$d(\vec{\beta} \wedge \vec{\Omega})^2 = -\frac{4}{h} (\vec{\beta} \wedge \vec{\Omega})^2 dz + 2D(\vec{\beta} \wedge \vec{\Omega})(d\vec{W} \wedge \vec{\Omega}) + 2D^2 \vec{\Omega}^2 dz. \quad (69)$$

Yet, it is $\vec{\Omega}^2$:

$$d(\vec{\Omega}^2) = 2\vec{\Omega} \vec{\beta} dz \quad (70)$$

and

$$d(\vec{\Omega} \vec{\beta}) = -\frac{2}{h} \vec{\Omega} \vec{\beta} dz + D\vec{\Omega} d\vec{W} + \vec{\beta}^2 dz. \quad (71)$$

Finally, the mean of $\vec{\beta}^2$ is known

$$\langle \vec{\beta}^2 \rangle = B^2. \quad (72)$$

Hence, the above set of equations is closed.

C. Solution of the Closed Set of Equation

Let us now integrate the above equations. This is clearly feasible, since each equation is linear. As $\langle \vec{\Omega} d\vec{W} \rangle = 0$ (i.e., the W_i are nonanticipating stochastic processes), we have

$$d\langle \vec{\Omega} \vec{\beta} \rangle = \frac{-2}{h} \langle \vec{\Omega} \vec{\beta} \rangle dz + \langle \vec{\beta} \rangle dz. \quad (73)$$

So with (65)

$$\langle \vec{\Omega} \vec{\beta} \rangle = \frac{B^2 h}{2} [1 - e^{-(2z/h)}] \quad (74)$$

and as in (64), we find

$$\langle \vec{\Omega}^2 \rangle = 2 \int_0^z \langle \vec{\Omega} \vec{\beta} \rangle dz' = \frac{B^2 h^2}{2} \left[2 \frac{z}{h} - 1 + e^{-2(z/h)} \right]. \quad (75)$$

Next, we also have

$$d\langle (\vec{\beta} \wedge \vec{\Omega})^2 \rangle = \frac{-4}{h} \langle (\vec{\beta} \wedge \vec{\Omega})^2 \rangle dz + 2D^2 \langle \vec{\Omega}^2 \rangle dz. \quad (76)$$

While $\langle (\vec{\beta} \wedge \vec{\Omega})(\vec{W} \wedge \vec{\Omega}) \rangle = 0$. Because of (65), then

$$\langle (\vec{\beta} \wedge \vec{\Omega})^2 \rangle = \frac{B^4 h^2}{6} \left[4e^{-2(z/h)} - 3 + 4 \frac{z}{h} - e^{-4(z/h)} \right]. \quad (77)$$

So

$$d\langle \vec{\Omega}_\omega(\vec{\beta} \wedge \vec{\Omega}) \rangle = \frac{-2}{h} \langle \vec{\Omega}_\omega(\vec{\beta} \wedge \vec{\Omega}) \rangle dz + \langle (\vec{\beta} \wedge \vec{\Omega})^2 \rangle dz. \quad (78)$$

Becomes with (65)

$$\langle \vec{\Omega}_\omega(\vec{\beta} \wedge \vec{\Omega}) \rangle = \frac{B^4 h^3}{12} \left[4 \frac{z}{h} - 5 + e^{-4(z/h)} + 4e^{-2(z/h)} + \frac{z}{h} e^{-2(z/h)} \right]. \quad (79)$$

And, finally, after a last integration we obtain the desired fiber length ℓ dependence

$$\begin{aligned} \langle \vec{\Omega}_\omega^2 \rangle &= \int_0^\ell 2\langle \vec{\Omega}_\omega(\vec{\beta} \wedge \vec{\Omega}) \rangle dz \\ &= \frac{B^4 h^4}{3} \left[\frac{17}{8} - \frac{5}{2} \frac{\ell}{h} + \frac{\ell^2}{h^2} - \frac{1}{8} e^{-4(\ell/h)} - 2e^{-2(\ell/h)} - 2 \frac{\ell}{h} e^{-2(\ell/h)} \right]. \end{aligned} \quad (80)$$

This is (23) of the main text.

APPENDIX D

DERIVATION OF THE PROBABILITY DISTRIBUTION WHEN ℓ IS LARGE COMPARED TO h

The basic equation that we shall need in this Appendix is the dynamical equation for PMD [6], [7] which relates the principal states to the fiber birefringence and length:

$$d\vec{\Omega}(z, \omega) = \vec{\beta}(z) dz + \omega \vec{\beta}(z) dz \wedge \vec{\Omega}(z, \omega) \quad (81)$$

where $\vec{\beta}(z)$ represents the local modal birefringence in [ps/km] at position z along the fiber. It fluctuates along the fiber with zero mean and correlation length h .

When the correlation length h goes to zero, the birefringence can be described by white noise. In that case, called case of long fibers $\ell \gg h$ the modal birefringence that appears in the dynamical equations for first and second-order PMD can be described by three independent Gaussian Markovian stochastic processes $\vec{W}$, called Wiener processes [16]

$$\lim_{h \rightarrow 0} \vec{\beta} dz = \sqrt{\frac{B^2 h}{3}} d\vec{W}. \quad (82)$$

It is well known that the stochastic equation one obtains in such a limit obeys Stratonovich calculus [16], that is the equation still remains in the same shape but with Stratonovich rules of calculus

$$d\vec{\Omega}(z, \omega) = \sqrt{\frac{B^2 h}{3}} d\vec{W} + \omega \sqrt{\frac{B^2 h}{3}} \vec{W} \circ \vec{\Omega}(z, \omega) \quad (83)$$

where $\wedge$ denotes the vector product and the little circle $\circ$ denotes the Stratonovich product. In order to compute mean values, it is convenient to rewrite (83) using Itô's stochastic calculus. To go from Stratonovich to Itô, we need to replace the Stratonovich Law $\circ$ by a development to first order

$$X \circ dY = X dY + \frac{1}{2} dX dY. \quad (84)$$

So that

$$d\vec{\Omega} = \sqrt{\frac{B^2 h}{3}} d\vec{W} + \omega \sqrt{\frac{B^2 h}{3}} d\vec{W} \wedge \vec{\Omega} + \frac{1}{2} \omega \sqrt{\frac{B^2 h}{3}} d\vec{W} \wedge d\vec{\Omega} \quad (85)$$

$$= \sqrt{\frac{B^2 h}{3}} d\vec{W} + \omega \sqrt{\frac{B^2 h}{3}} d\vec{W} \wedge \vec{\Omega} - \omega^2 \frac{B^2 h}{3} \vec{\Omega} dz. \quad (86)$$

Because

$$d\vec{W} \wedge d\vec{\Omega} = \sqrt{\frac{B^2 h}{3}} d\vec{W} \wedge (d\vec{W} \wedge \vec{\Omega}) \quad (87)$$

$$= \sqrt{\frac{B^2 h}{3}} d\vec{W} (d\vec{W} \vec{\Omega}) - \vec{\Omega} (d\vec{W}^2) \quad (88)$$

$$= \sqrt{\frac{B^2 h}{3}} (\vec{\Omega} dz - \vec{\Omega} 3dz) \quad (89)$$

$$= -2 \sqrt{\frac{B^2 h}{3}} \vec{\Omega} dz. \quad (90)$$

Accordingly, for long fibers, the dynamical equations for first- and second-order PMD, i.e., the evolution equation for $\vec{\Omega}$ and its derivative versus fiber length simplify then to

$$d\vec{\Omega} = \sqrt{\frac{B^2 h}{3}} d\vec{W} + \omega \sqrt{\frac{B^2 h}{3}} d\vec{W} \wedge \vec{\Omega} - \omega^2 \frac{B^2 h}{3} \vec{\Omega} dz \quad (91)$$

$$d\vec{\Omega}_\omega = \sqrt{\frac{B^2 h}{3}} d\vec{W} \wedge \vec{\Omega} + \omega \sqrt{\frac{B^2 h}{3}} d\vec{W} \wedge \vec{\Omega}_\omega - 2\omega \frac{B^2 h}{3} \vec{\Omega} dz - \omega^2 \frac{B^2 h}{3} \vec{\Omega}_\omega dz. \quad (92)$$

The noise $\vec{W}$ is independent of the position z along the fiber and of the angular frequency ω of the signal. At the entrance of the fiber, the initial conditions are

$$\vec{\Omega}(z=0, \omega) = \vec{0} \quad \vec{\Omega}_\omega(z=0, \omega) = \vec{0}. \quad (93)$$

Dealing with those equations, it is a great property that the moments of both $\vec{\Omega}$ and $\vec{\Omega}_\omega$ do not depend on the angular frequency, as shown in Appendix C. It is thus possible to use simplified equations to derive the moments and distribution functions of $\vec{\Omega}$ and $\vec{\Omega}_\omega$. Let us set $\omega = 0$

$$d\vec{\Omega} = \sqrt{\frac{B^2 h}{3}} d\vec{W} \quad (94)$$

$$d\vec{\Omega}_\omega = \sqrt{\frac{B^2 h}{3}} d\vec{W} \wedge \vec{\Omega} \quad (95)$$

where at the entrance of the fiber, the initial conditions remain.

Finally, we obtain $\vec{\Omega} = (\sqrt{B^2 h/3}) \vec{W}$ and

$$d\vec{\Omega}_\omega = \frac{B^2 h}{3} d\vec{W} \wedge \vec{W}. \quad (96)$$

This equation is well known in the mathematical literature under the name of ‘‘Levy Stochastic Area.’’ The characteristic function and cumulants are found in [17]. With a table of Fourier transforms [18], the probability distribution $\hat{\rho}_\ell$ for each component $k = x, y, z$ of the vector $\vec{\Omega}_\omega$ can be computed as a function of the fiber parameters B, h , and length ℓ :

$$\hat{\rho}_\ell(\Omega_{\omega,k}) = \frac{3}{2B^2 h \ell \cosh\left(\frac{3\pi}{2B^2 h} \frac{\Omega_{\omega,k}}{\ell}\right)}. \quad (97)$$

Thanks to spherical symmetry, the probability distribution for the magnitude $\Omega_\omega = |\vec{\Omega}_\omega|$ can easily be derived:

$$\rho_\ell(\Omega_\omega) = -2\Omega_\omega \frac{d\hat{\rho}_\ell(\Omega_\omega)}{d\Omega_\omega}. \quad (98)$$

Hence, we obtain the probability distribution of the magnitude $\Omega_\omega = |\vec{\Omega}_\omega|$ in the case of long fiber

$$\rho_\ell(\Omega_\omega) = \frac{2c^2 \Omega_\omega \sinh(c\Omega_\omega/\ell)}{\pi \ell^2 \cosh^2(c\Omega_\omega/\ell)} \quad (99)$$

where c is the following function of the second-order PMD coefficient:

$$c \equiv \frac{3\pi}{2B^2 h} = \frac{\sqrt{3}\pi u}{2\text{PMD}_{\text{coeff}}^{\text{second-order}}}. \quad (100)$$

From this distribution, one can deduce all momenta of Ω_ω as well as the probability of instantaneous values to be larger than any given upper bound.

APPENDIX E

SUMMARY OF SOME GENERAL RELATIONS

Equation (101) summarizes some general relations between various combinations of the first- and second-order PMD vector $\vec{\Omega}$ and $\vec{\Omega}_\omega$ to the fiber length ℓ . These relations hold in the case of long fibers: $\ell > h$. For simplicity we are using

normalized units; in order to recover the physical units, the coordinate z that appears in this Appendix should be replaced by $z \rightarrow B^2 h \ell / 3$.

$$\left[\begin{array}{ll} \langle \vec{\Omega} \rangle = \vec{0} & \langle \vec{\Omega}^2 \rangle = 3z \\ \langle \vec{\Omega}_\omega \rangle = \vec{0} & \langle \vec{\Omega}_\omega^2 \rangle = 3z^2 \\ \langle \vec{\Omega} \vec{\Omega}_\omega \rangle = \vec{0} & \langle (\vec{\Omega} \vec{\Omega}_\omega)^2 \rangle = z^3 \\ \langle \vec{\Omega}^4 \rangle = 15z^2 & \langle \vec{\Omega}_\omega^4 \rangle = 25z^4 \\ \langle \vec{\Omega}^2 \vec{\Omega}_\omega^2 \rangle = 13z^3 & \\ \langle \vec{\Omega} \wedge \vec{\Omega}_\omega \rangle = \vec{0} & \langle (\vec{\Omega} \wedge \vec{\Omega}_\omega)^2 \rangle = 12z^3 \end{array} \right]. \quad (101)$$

Equation (102) lists various relations involving components of the first- and second-order PMD vector $\vec{\Omega}$ and $\vec{\Omega}_\omega$ that were proposed in [8] based on numerical evidence. We could prove them under the assumption that they are independent of the optical frequency ω (we omit the proofs as they are lengthy and quite similar to the ones already presented in this article).

$$\left[\begin{array}{l} \forall i, j, \langle \Omega_i \Omega_{\omega j} \rangle = 0 \\ \forall i \neq j, \langle \Omega_{\omega i} \Omega_{\omega j} \rangle = 0 \\ \forall i \neq j, \langle \Omega_i^2 \Omega_{\omega j}^2 \rangle = \frac{5}{3} \langle \Omega_i^2 \rangle \langle \Omega_{\omega j}^2 \rangle \\ \forall i, \langle \Omega_i^2 \Omega_{\omega i}^2 \rangle = \langle \Omega_i^2 \rangle \langle \Omega_{\omega i}^2 \rangle \\ \forall i \neq j, \langle \Omega_{\omega i}^2 \Omega_{\omega j}^2 \rangle = \frac{5}{3} \langle \Omega_{\omega i}^2 \rangle \langle \Omega_{\omega j}^2 \rangle \end{array} \right]. \quad (102)$$

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F. Prieto, photograph and biography not available at the time of publication.